

Machine Learning Basics

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Machine learning is a way for computers to learn patterns from examples. Instead of writing one rule for every situation, we give an algorithm data and ask it to build a model. The model then uses what it learned to make predictions about new data.

A simple workflow

A typical project starts with a question, such as predicting whether a student will complete a course. We collect relevant examples, clean the data, choose useful features, and split the examples into training and test sets. The algorithm learns from the training set. The test set is kept separate so that we can measure how well the model works on unseen examples.

Supervised and unsupervised learning

In supervised learning, each training example includes a target answer. Classification predicts a category, while regression predicts a number. In unsupervised learning, the data has no target answer. The algorithm may group similar examples or find a smaller set of patterns that summarizes the data.

Important terms

A feature is an input used by the model. A label is the answer in supervised learning. Training means adjusting the model from examples. Inference means using the trained model to produce a prediction. Generalization is the ability to perform well on data that was not used during training.

Retrieval check: What does BASIC-ORCHID-17 say about the difference between training and inference?

Overfitting

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Overfitting happens when a model learns the training data too closely, including noise and accidental details. It may score very well on examples it has already seen but perform poorly on new examples. This gap is a warning that the model has memorized more than it has generalized.

How to recognize it

Compare performance on training data and validation data. If training error keeps falling while validation error begins to rise, overfitting is likely. A large gap between training and validation performance is another common signal. The final test set should not be used repeatedly during model selection because that would leak information into the process.

Why it happens

Overfitting is more likely when the model is very flexible, the dataset is small, labels contain mistakes, or training continues for too long. Irrelevant features can also give the model opportunities to learn coincidences that do not repeat in the real world.

Practical responses

Collect more representative data when possible. Simplify the model, remove unreliable features, use cross-validation, stop training earlier, or apply regularization. These methods reduce the model's ability to depend on fragile details.

Retrieval check: According to OVERFIT-CEDAR-29, what pattern in training and validation error suggests overfitting?

Regularization

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Regularization is a set of techniques that discourages a model from becoming more complex than necessary. The goal is not to make the training score perfect. The goal is to learn a stable pattern that continues to work on unseen data.

Penalty-based methods

L1 regularization adds a cost based on the absolute size of model weights. It can push some weights to exactly zero, which may create a sparse model. L2 regularization adds a cost based on squared weights. It usually makes weights smaller and spreads influence across features instead of allowing a few very large weights.

Other forms

Early stopping ends training when validation performance stops improving. Dropout randomly removes some neural network units during training, which reduces dependence on a single path. Data augmentation creates reasonable variations of training examples, helping the model learn patterns that remain stable under those changes.

Choosing the strength

Regularization strength is a hyperparameter. Too little may leave the model overfit. Too much can make the model too simple and cause underfitting. Choose the value with a validation set or cross-validation, then report final performance once on a separate test set.

Retrieval check: In REGULAR-MAPLE-43, how do L1 and L2 regularization affect weights differently?

Bias-Variance Tradeoff

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The bias-variance tradeoff explains two different ways a model can make errors. Bias comes from assumptions that are too simple. Variance comes from being too sensitive to the exact training sample. Good generalization usually requires a balance between them.

High bias

A high-bias model misses important structure in both training and new data. This is underfitting. Training error and validation error are often both high and fairly close. Useful responses include adding informative features, increasing model capacity, or reducing excessive regularization.

High variance

A high-variance model changes too much when trained on a different sample. It often has low training error but noticeably higher validation error. Useful responses include collecting more data, simplifying the model, increasing regularization, or averaging several models.

Finding a useful balance

Model complexity often moves bias and variance in opposite directions. Increasing complexity can reduce bias but increase variance. Validation results help locate a practical middle point. The best model is not automatically the most complex one; it is the one that performs reliably on unseen data for the intended task.

Retrieval check: Using TRADEOFF-RIVER-61, compare the error patterns of high bias and high variance.